

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru | github.com/zac-09 | Kampala, Uganda (EAT, remote)

PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years building Node.js and TypeScript services, with deep MongoDB experience across migration, schema design, and live-traffic sync. Led a zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP at MTailor, reporting directly to the CTO, with cost savings of \$5K/month. Built an event-driven two-way MongoDB ↔ Firestore sync using Node.js + Google Pub/Sub with idempotent consumers — the kind of durability and correctness work MongoDB at scale demands. Now focused on senior backend infrastructure roles where reliability, performance, and technical direction meet.

CORE COMPETENCIES

MongoDB (migration, schema, indexing)

Node.js / TypeScript

Distributed Systems

Event-Driven Architecture

Zero-Downtime Migrations

Reliability & Performance

Microservices (Kafka, K8s)

Technical Direction

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer (US Remote) — Backend, Data Infrastructure

- Led end-to-end migration of 20+ production applications off **Parse/MongoDB** to Firebase/GCP with zero downtime, reporting directly to the CTO.
- Designed and operated a real-time two-way **MongoDB ↔ Firestore sync** using Node.js + Google Pub/Sub with idempotent consumers, retries, and back-pressure handling.
- Migrated file storage from S3 to GCS via a Python script, preserving referential integrity across millions of objects.
- Shipped Express Shipping feature (\$40 additional revenue per order) end-to-end.
- Saved the company \$5,000/month by migrating all services off AWS.

MTailor (Contractor)

Mar 2022 – Jul 2022

Contractor — MongoDB → Firestore Migration

- Built a schema-preserving migration pipeline reading **MongoDB** and writing to Firestore with complex transformation logic.
- Authored Firebase SDK documentation (Node.js + Python) for the engineering team.
- Trained the Ops team on the new Firebase dashboard and backup procedures.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer (Singapore Remote)

- Built ad-hoc jobs querying **over 2 million Firestore records** to keep customer prices current from Morningstar APIs.
- Extended backend APIs with Firebase Cloud Functions and Express.js.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead — Legal Tech for NGOs

- Led a team of 4 developers delivering case management + paralegal database systems for FIDA Uganda and LASPNET.
- Architected a microservices backend using **Apache Kafka, Docker, Kubernetes** with gRPC inter-service communication.

PROJECTS

Real-time MongoDB ↔ Firestore Sync Pipeline (MTailor)

Live bidirectional sync supporting parallel production traffic during a 20+ app migration. Idempotent Pub/Sub consumers, schema translation, retry policy.

Node.js · MongoDB · GCP Pub/Sub · Firestore

MongoDB → Firestore Migration Tooling

Python + Node.js pipeline that read MongoDB collections, applied complex processing, and wrote to Firestore without service interruption.

Python · Node.js · MongoDB · Firestore

FIDA Uganda Case Management — Microservices Backend

Kafka + Docker + Kubernetes microservices for a multi-actor NGO case management platform; gRPC between services.

Node.js · Kafka · Docker · Kubernetes · gRPC

EDUCATION

B.Sc. Computer Engineering — Makerere University

Aug 2018 – Sep 2022

SKILLS

Proficient: MongoDB, Node.js, TypeScript, Firebase, GCP (Firestore, Pub/Sub, GCS, Cloud Functions), Python, SQL, React, Git, Linux

Intermediate: Go, AWS (S3, EC2, EBS), Rust, React Native

Patterns: Distributed systems, event-driven architecture, zero-downtime migrations, microservices, reliability engineering

Languages: English · Kiswahili